Network modelling - Assignmen 1 - Group 24

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Task 1

TODO

Task 2

TODO: Think about whether to use one-sided or two sided tests to report significance

(1)

After running the QAP logistic regression, we see a slope coefficient of 2.93 with a p-value of 0.00. Therefore the result is significant and we can state that friendship in W1 is positively correlated with friendship in W2.

(2)

The values are defined as follows:

```
literacyReceiver <- matrix(attr$literacy_end, length(ids), length(ids), byrow = TRUE) # (i)
sameGender <- outer(attr$gender, attr$gender, "==") * 1 # (ii)
hiseiSender <- matrix(attr$HISEI, length(ids), length(ids), byrow = FALSE) # (iii)
```

- For (i) we create a matrix where each column is filled with the literacy score of the corresponding student.
- For (ii) we create a matrix with one in case of a gender match, and zero otherwise.
- For (iii) we create a matrix where each row is filled with the HISEI score of each student

(3)

Here are the coefficients for the regressors together with the corresponding p-values

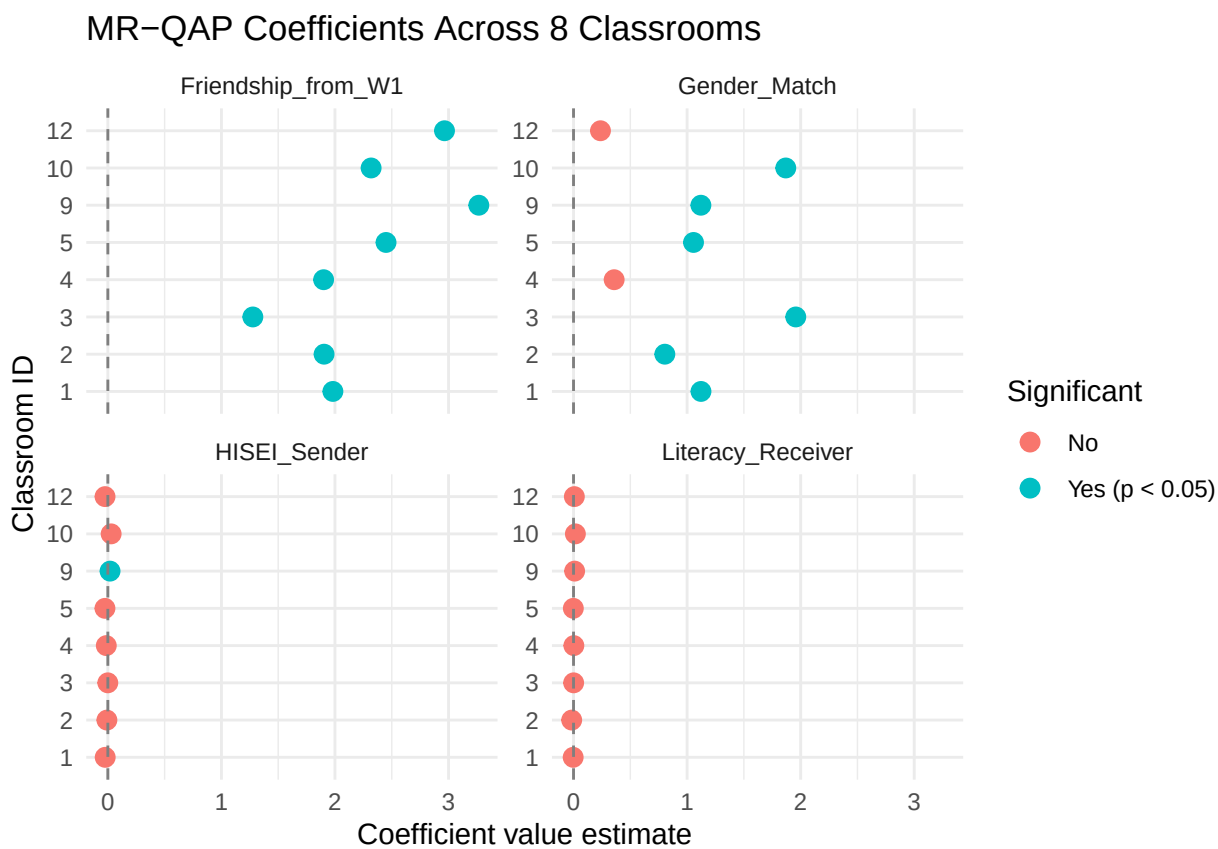
Name	Coefficient	p-value
Friendship W1	2.318175	0.0000
Literacy receiver	0.0163386306	0.1282
Same gender	1.8688700	0.0000
HISEI sender	0.0290319336	0.1752

Here are the corresponding interpretations (for the 5% significance level):

- The coefficient from (i) is insignificant, therefore we cannot confirm that higher literacy scores are associated with more friendship nominations received.
- The coefficient from (ii) is positive and significant, so we can confirm the statement that having the same gender increases the likelihood of having a friendship nomination (while everything else stays the same).
- The coefficient from (iii) is insignificant, so we cannot confirm the statement that higher HISEI values are associated with more friendship nominations send.

Note: Insignificant coefficients mean that we cannot reject the null hypothesis that there is no correlation (the coefficient is 0).

(4)



We can here see the values for all coefficient across the different classrooms.

(5)

- (1) Friendship from W1 - The results generalize pretty with the coefficients staying significant and positive across all classrooms
- (2.i) Literacy receiver - The results are insignificant across all classrooms
- (2.ii) Gender match - The results are significant across almost all classrooms and the coefficients are always positive. This suggest that our conclusion (that having the same gender is positively correlated with friendship in W2) should generalize.
- (2.iii) HISEI sender - The results are almost always insignificant. Whenever they are significant, they are very close to zero. We can't say for sure whether the hypothesis is true or not.

Task 3

(1)

We can see that the network correlation coefficient is estimated to be around 0.85 with a p-value of 0, which makes it significant. Therefore, only taking W2 friendship into account, we can say that students tend to mark friends people with similar literacy scores.

(2) & (3)

When we run the model with the additional variables, we get the following estimates:

Table 2: Extracted Results for Classroom 10

Variable	Estimate	P_Value
Gender (Female)	20.8824892	0.0046561
HISEI	0.9727459	0.0000397
Network Autocorrelation (rho)	-0.9395290	0.0282669

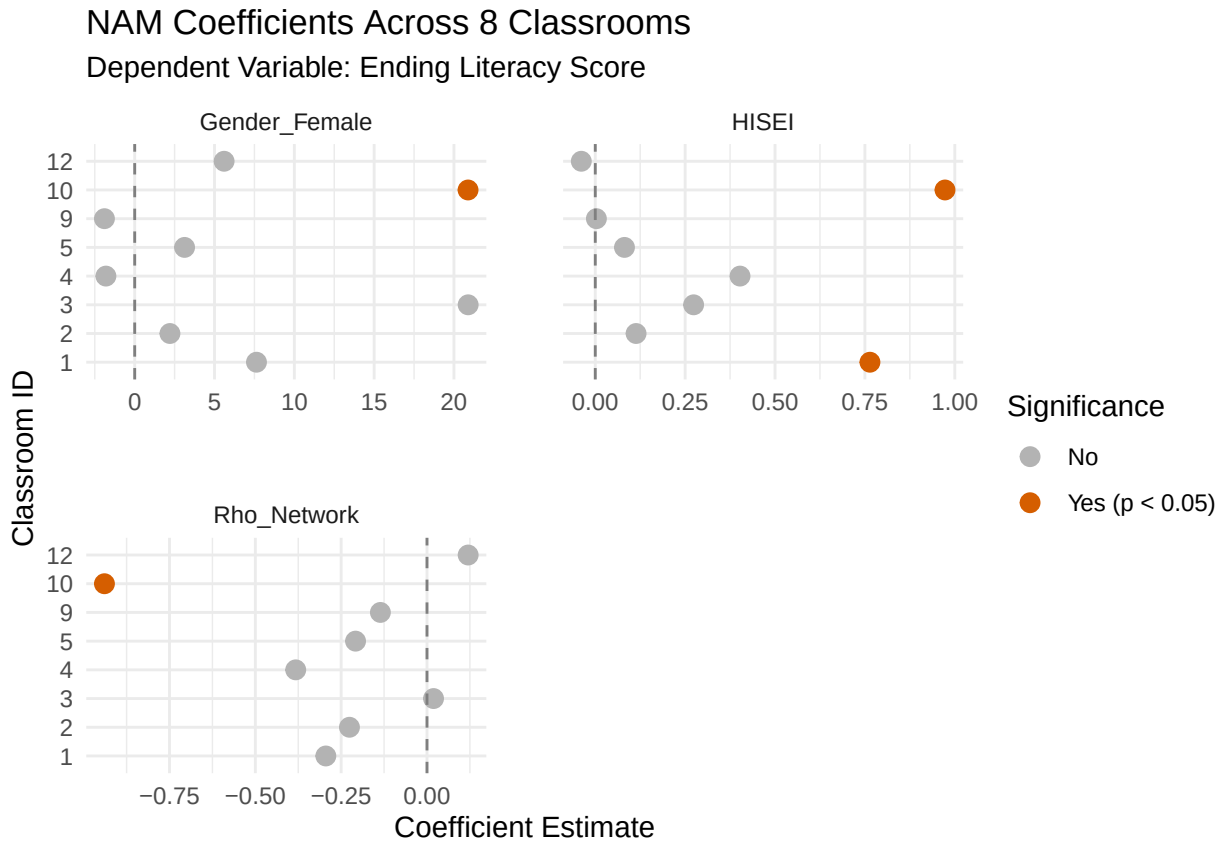
We can see that all the coefficients are significant at the 5% significance level, therefore we can interpret them like so:

- Being a female (all other things staying the same) leads to a better literacy score. The estimated increase is around 20.88.
- Having a higher HISEI score (all other things staying the same) leads to a better literacy score.
- The network autocorrelation coefficient is now negative. In (1) it was estimated to be positive. This means that in the current model it is observed that people tend to make friends with others with different literacy score (higher or lower). The autocorrelation hypothesis is therefore suggested by the data.

(4)

TODO

(5)



We can see that our results do not look very generalizable and that classroom 10 turns out to be an outlier. Observing the other classrooms, we can see that almost every time the coefficients are non-significant, which means that we cannot draw the same conclusions for other classrooms.